



UNIVERSITY INTERNSHIP REPORT

Web Development Internship

Social Media and E-Commerce Platforms

INTERNSHIP DURATION

June 2026 – August 2026 (8 Weeks)

STUDENT CREDENTIALS

Abir Ekansh

Enrollment / Roll No.
25SCS1003005283

Section & Program
2CSE31 | B.Tech (CSE)

HOST ORGANIZATION

Codec Technologies

Guided & Deployed Open-Source Frameworks

Candidate & Program Overview

- **Student Name:** Abir Ekansh
- **Roll Number:** 25SCS1003005283
- **Section:** 2CSE31
- **Degree:** B.Tech (Computer Science & Engineering)
- **University:** IILM University, Greater Noida, U.P.
- **Primary Focus:** Full-Stack Web Development
- **Core Deliverables:** Social Media App & E-Commerce Platform



The Industry Challenge & Problem Statement



Fragmented Engagement

In the digital age, users require seamless, high-performance platforms to communicate and share media without relying on bloated, generalized legacy solutions.



Retail Inefficiencies

Existing e-commerce frameworks are often overly complex to scale. There is a strong need for streamlined online shopping, optimized cart states, and secure checkout experiences.



Secure Architectures

The primary engineering challenge lies in architecting systems from scratch that handle robust user authentication, real-time data flow, and secure state management flawlessly.

Core Project Objectives



Social Network Implementation: Construct a scalable Social Application allowing users to seamlessly create profiles, authenticate securely, post updates, and interact.



E-Commerce System Development: Develop a robust E-Commerce system featuring a dynamic product catalog, state-managed shopping cart functionality, and simulated secure checkout.



Responsive User Interfaces: Implement modern, highly responsive frontend interfaces that provide optimal and engaging UI/UX experiences across varying device screen sizes.



Robust Backend Infrastructure: Design and deploy backend RESTful APIs capable of secure database CRUD operations and efficient data retrieval pipelines.

Technologies & Tools Used



Frontend Development

Utilized HTML5, CSS3, JavaScript, and React.js to build dynamic, component-based user interfaces and manage complex application states effectively on the client side.



Backend Architecture

Leveraged Node.js and Express.js to create a structured server environment, enforce strict business logic, and process JSON Web Tokens (JWT) securely.



Database & Deployment

Implemented MongoDB/SQL for relational and document data management, while utilizing Git and GitHub for continuous version control and project hosting.

System Architecture

Client-Server Model

Both applications leverage a strict Client-Server architecture to separate concerns, improve security, and enhance scalability.

The React-based frontend layers handle data presentation and state management, communicating asynchronously with Node.js REST APIs. The backend services enforce business logic, process tokens, and execute CRUD operations securely against the database clusters.



Agile Development Methodology

Sprint 2 - Backend APIs

Building, securing, and rigorously testing backend REST API endpoints and core server logic operations.

Sprint 4 - Deployment

Thorough full-stack testing, debugging, repository finalization, and deployment of both web applications.

Sprint 1 - Planning

Initial focus on gathering strict requirements and mapping out logical database schemas for users and products.

Sprint 3 - UI/UX

Iterative development and integration of interactive React frontend components based on backend functionality.

Expected Outcomes & Deliverables



Social Platform

A fully functional social app demonstrating clear proficiency in managing relational user data, secure authentication, and dynamic social feeds.



E-Commerce System

A digital storefront showcasing the ability to handle complex transactional logic, product states, and seamless cart session management.



Robust Deployment

Two scalable digital platforms built from scratch utilizing modern web development architectures, available on GitHub.



CERTIFICATE OF INTERNSHIP

This certificate is awarded to
Abir Ekansh

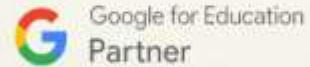
From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 1 Month AICTE & ICAC

Approved internship program as

Web Developer Intern

Conducted from 25/07/2026 to 25/08/2026



AICTE ID - CORPORATE175A2N4Ruc5hr1732940333

A handwritten signature in black ink, appearing to read "Anurag", positioned above a horizontal line.

Dr. Anurag Shrivastava
Program Manager

Thank You!

End of Presentation

 github.com/Abir-Ekansh